

Week 4 Evolutionary algorithm 2 solutions

1. What is selection pressure? How can we cause low pressure and how can we cause high pressure?

Selection pressure is the pressure to be the best individual in a population. We can have a high pressure by always selecting the best individuals. This pushes the population to be as strong as possible. We can have a lower pressure by allowing weaker individuals to stay in the population. This generates less fit individuals.

2. Why do we not always want high selection pressure?

We do not always want the most fit individuals to stay in a population, because this can cause us to be stuck in a local optima. By allowing less fit individuals to stay, we can explore more and eventually find global optima.

3. Parent selection: what is fitness-proportionate selection, what is the problem with this type of selection? What is rank-based selection and how does this effect selection pressure? What is tournament selection?

Fitness-proportionate selection is when we choose the parents based on their fitness-probability. The higher their fitness is, the more likely they are to be chosen as a parent. The problem is that it almost always chooses the individuals with higher fitness, not allowing more exploration. This can make high pressure early on, and make individuals become too similar.

Rank-based selection rank the individuals from strongest to weakest, then base selection probabilities on rank. The selection pressure is the same.

Tournament is when we choose k individuals and choose the most fit out of these k . we do this several times until we have enough parents

4. Name some ways to choose the survivor selection? (3 ways)

Elitism is when we choose the n best individuals, then fill the rest randomly.

Another method is to always choose the best amongst the children. This can risk that we lose the best solution, because we do not consider the old generation (parents)

The last method is to choose the best out of the old population and the newly generated children.

5. Diversity preservation: What is the difference between implicit and explicit approaches for preserving diversity?

Both are methods to preserve diversity. Explicit explicitly changes the probability of an individual being selected to the new generation. Less likely to select those that are similar, and more likely to choose those that are different.

Implicit preserve the diversity by making changes to structure.

Explicit approaches usually make the individuals that are similar compete for fitness by explicitly changing the probability that they are selected by selection operators. Explicit methods include fitness sharing and crowding.

Implicit approaches usually restrict the population into groups to try to preserve the diversity between the groups, but do not explicitly guarantee that solutions that are similar to others are given a disadvantage during selection. Implicit methods include automatic speciation and geographical separation

6. What is crowding and fitness sharing in explicit approach?

Fitness sharing is when similar solutions share the total fitness value. We want to restrict the individuals within a given niche by sharing fitness.

fitness sharing addresses the diversity by adjusting the fitness of individuals based not only on their own performance but also on the performance of other individuals in the population. Specifically, it **reduces the fitness of an individual based on the similarity of its genome to those of other individuals in the population**. This reduces the selection pressure on individuals that are similar to others and promotes the preservation of diversity.

Fitness sharing can be particularly useful in problems where there are multiple equally valid solutions, or where there are multiple subproblems that need to be solved simultaneously. By promoting diversity, it can help genetic algorithms explore a wider range of solutions and avoid getting stuck in local optima.

If the solutions are different, $sh(d)$ should be 0, if the solutions are similar then the $sh(d)$ should be closer to 1

$$f'(i) = \frac{f(i)}{\sum_{j=1}^{\mu} sh(d(i, j))} \quad sh(d) = \begin{cases} 1 - d/\sigma & d \leq \sigma \\ 0 & otherwise \end{cases}$$

Crowding works by measuring the similarity between individuals in the population based on their genetic material, and ensuring that **individuals with similar genetic material are not allowed to occupy the same region of the search space**. In essence, it promotes diversity by encouraging individuals that are different from each other to occupy different regions of the search space.

Crowding is usually implemented by maintaining a fixed population size, **and replacing individuals that have been selected for reproduction with new offspring that have genetic material different from the individuals that were replaced**. This ensures that the population continues to explore different regions of the search space and avoids getting stuck in local optima.

New individuals replace similar individuals. We select parents and randomly pair two parents. Each offspring competes with their nearest parent for survival.

Crowding

- New individuals replace similar individuals
 - Each offspring competes with their nearest parent
- 7.** What is speciation and geographical separation in implicit approach?

Speciation is when we only allow similar individuals to reproduce with each other. Another way is to put tags to genotypes and when selecting parents to mate, we only pick members with a good match.

Geographical separation is when we have several EA's occurring separated from each other. Each come up with their own solution and every k generations, we send individuals between them to push evolution forwards. Island model.

- 8.** What is hybridization and why do we do it?

Hybridization is when we combine EA with another search method. We do this to improve EA, by combining we can more efficiently find the best solutions.

- 9.** What is memetic algorithm? How can we apply it to EA?

Memetic algorithm is when we combine local search operators in EA. This makes EA faster and more accurate. We can make small, but intelligent, changes to existing solution. We can do hill climbing on every individual per generation before applying variation operators.

- 10.** What is Lamarckian and Baldwinian in memetic algorithm?

Lamarckian is when we take an individual I, do local search to find a better solution I' and put I' back in the population and remove I. Baldwinian take individual I, do local search to find I' and calculate the fitness and set I fitness to I' fitness. The solution I is not replaced and I' is not put in population.

- 11.** What is multi-objective optimisation problems?

Multi-objective optimization problems is when we have more objectives that are conflicting. For example when we buy a car we have to evaluate speed over price over reliability.

- 12.** What two approaches do we have in MOP?

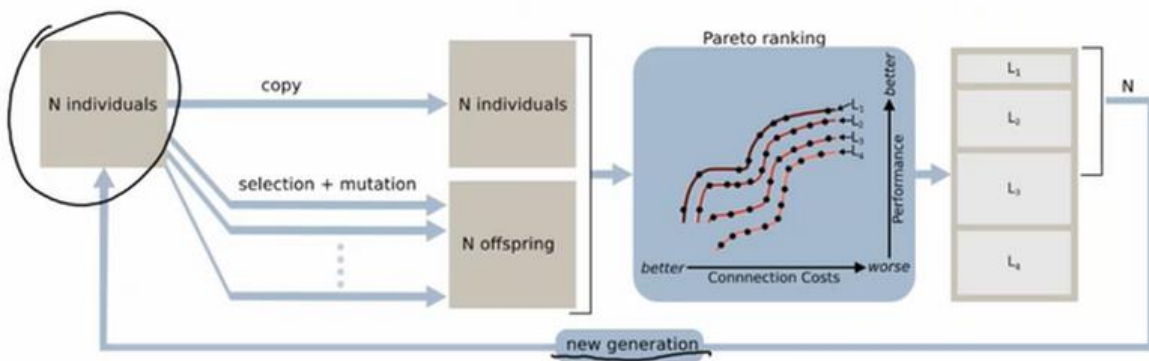
Weighted sum (scalarisation): transform into a single objective optimization method and compute a weighted sum of the different objectives.

A set of multi-objective solution (parent front): the population based nature of EAs used to simultaneously search for a set of points approximating parent front.

Bruger EA til å finne løsninger på pareto front

13. What is pareto front?

Parent front is a set of non-dominating solutions from the entire feasible solution space.



What is the goal of multi-objective optimizers?

Find a set of non-dominated solutions following the criteria of:

- Convergence (as close as possible to the pareto-optimal front)
- Diversity (spread, distribution)

Solutions on the Pareto optimal set have to be non-dominated. That is, no other solutions should be better according to both objectives (or even better on some but equally good on others).